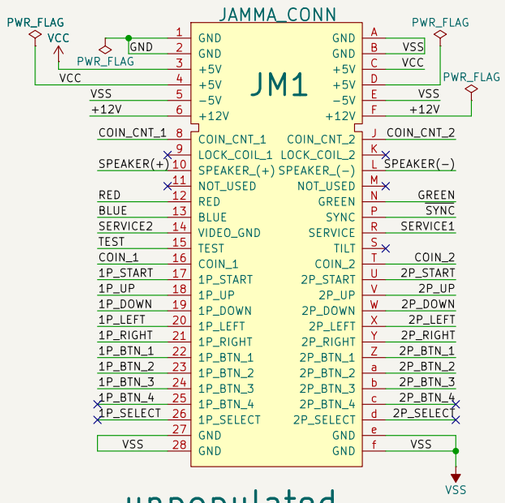
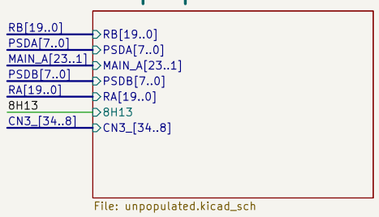
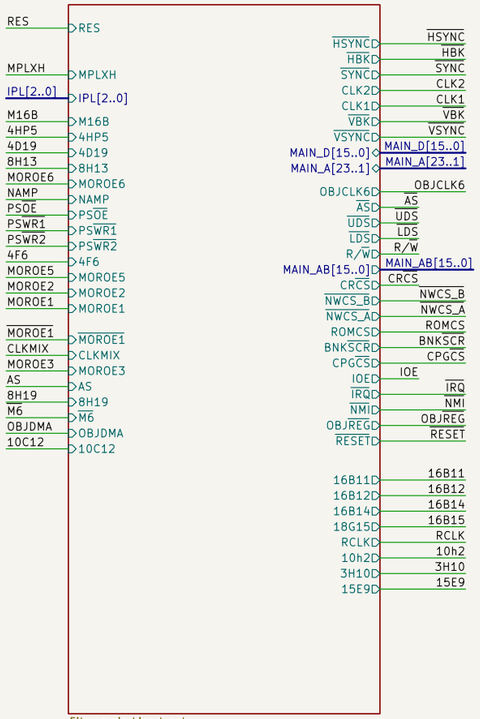


unpopulated



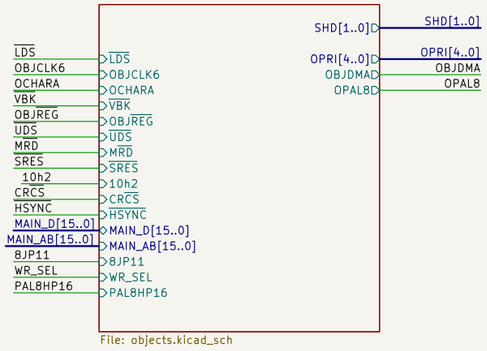
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main



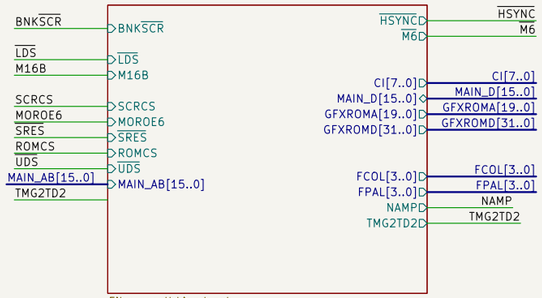
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objects



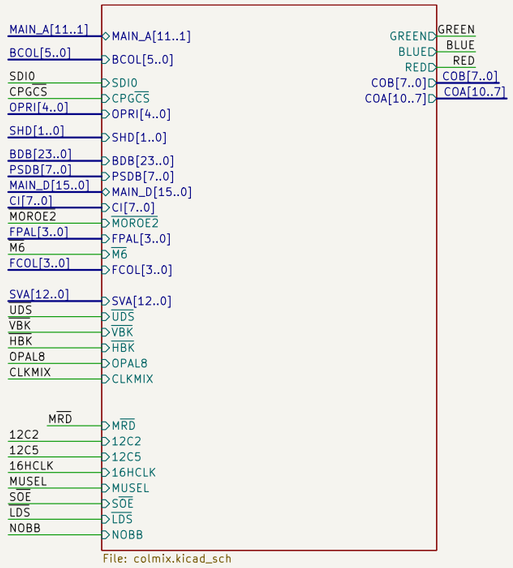
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scroll



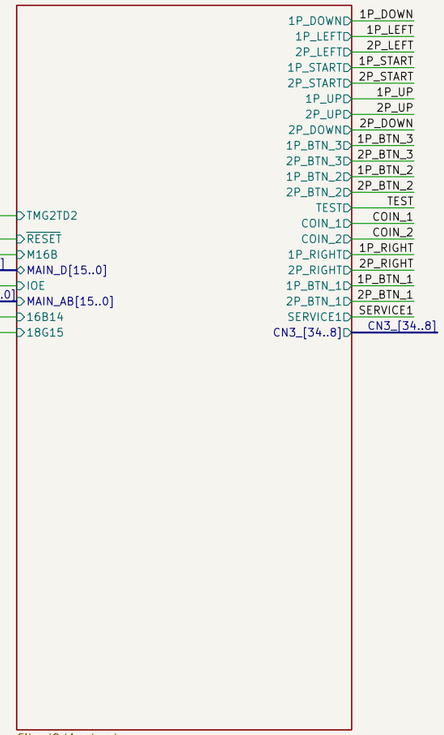
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colmix



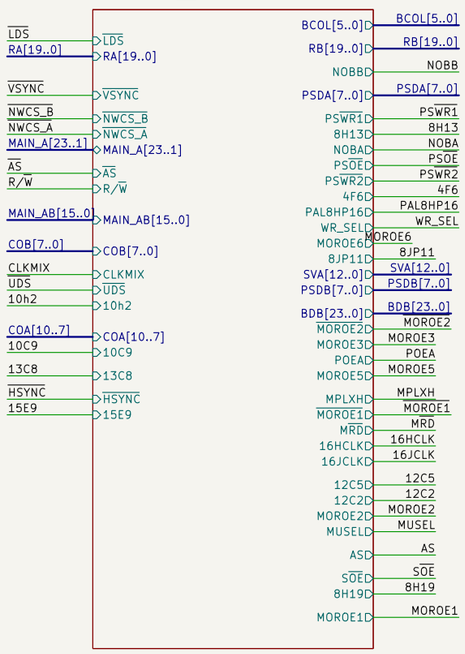
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IO



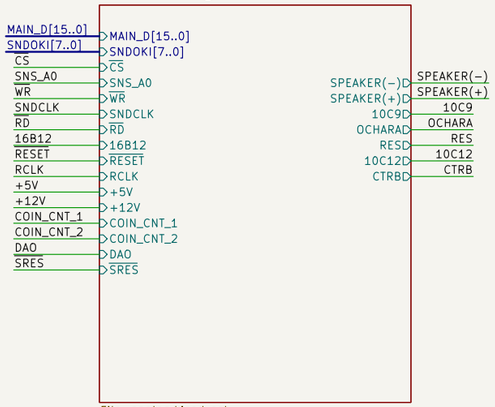
File: IO.kicad_sch

MAIN_PSAC



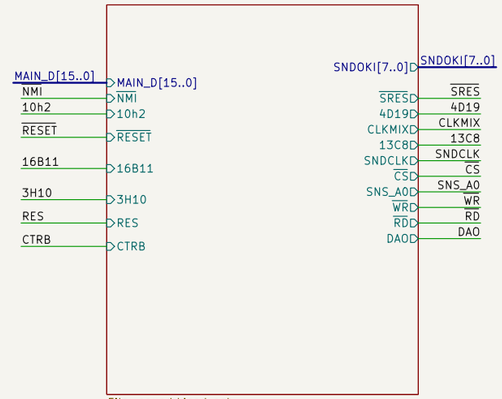
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analog



File: analog.kicad_sch

sound

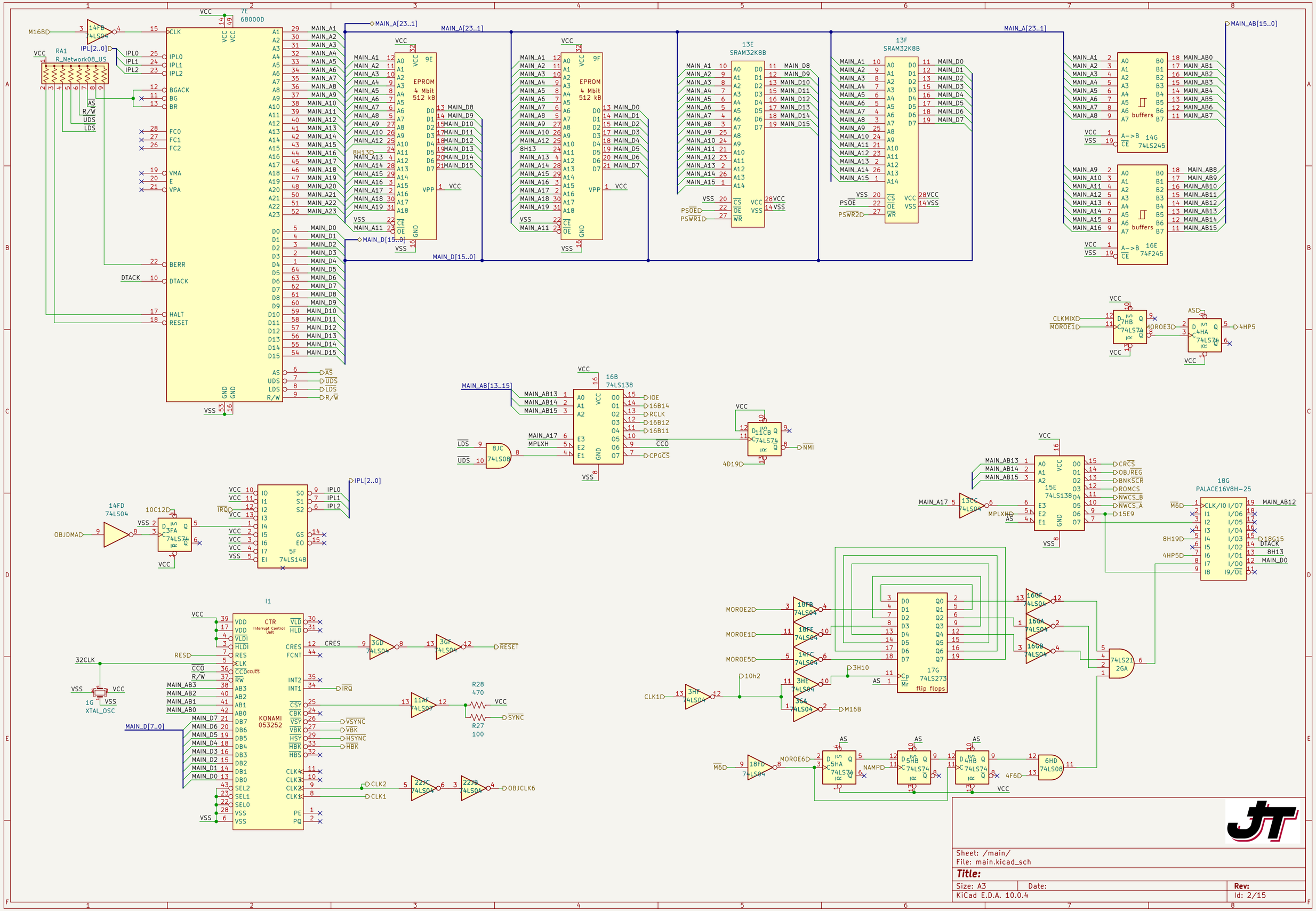


File: sound.kicad_sch

Sheet: /
File: dbz.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 1/15


7E
68000D
M16B
14FB
74LS04
IPL[2..0]
RA1
R_Network08_US
AS
R/W
UDS
LDS
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
MAIN_A[23..1]
MAIN_D[15..0]
9E
EPROM
4 Mbit
512 kB
9F
13E
SRAM32K8B
13F
SRAM32K8B
MAIN_AB[15..0]
buffers
14G
74LS245
16E
74F245
CLKMIX
MOROE1
MOROE3D
7HB
74LS74
4HA
4HP5
16B
74LS138
MAIN_AB[13..15]
IOE
16B14
RCLK
16B12
16B11
CCO
CPGCS
8JC
74LS08
NMI
4D19
11CB
10C12
14FD
74LS04
OBJDMAD
3FA
IPL0
IPL1
IPL2
5F
74LS148
13CC
MAIN_A17
MPLXH
CRCS
OBJREG
BNKSCR
ROMCS
NWCS_B
NWCS_A
15E9
18G
PALACE16V8H-25
M6D
8H19
4HP5
18G15
DTACK
8H13
MAIN_D0
I1
KONAMI
053252
Interrupt Control Unit
32CLK
1G
XTAL_OSC
RESD
CCO
R/W
MAIN_AB3
MAIN_AB2
MAIN_AB1
MAIN_AB0
MAIN_D[7..0]
CRES
IRQ
VSYNC
VBK
HSYNC
HBK
CLK2
CLK1
OBJCLK6
RESET
SYNC
R28
470
R27
100
MOROE2
MOROE1
MOROE5
MOROE6D
CLK1D
10h2
3H10
M16B
NAMPD
4F6D
17G
74LS273
flip flops
74LS21
2GA
6HD
Sheet: /main/
File: main.kicad_sch
Title:
Size: A3
Date:
KiCad E.D.A. 10.0.4
Rev:
Id: 2/15

OBJROMA[0..19]

MAIN_AB[15..0]

MAIN_D[15..0]

K3 KONAMI 053247 OBJECT DRAWING

TILE DRAWING

TILE DRAWING CLK

COLOR PALETTE GENERATOR

OBJECT ROM DATA BUS

GENERATOR TO DRAWING

RUNWAY OUTPUT NOT VISIBLE

OPRI[4..0]

OBJROMD'[15..0]

SHD[1..0]

G2D[21..0]

K4 KONAMI 053246 OBJECT GENERATOR

See Konami's Racing Force schematics for symbol reference

OBJECT ROM ADDRESS BUS

ROM CHECK

GENERATOR TO DRAWING

CLKOBJ'

LUTD[15..0]

EN[0..7]

EA[5..1]

XOBJROMD[31..0]

YOBJROMD[31..0]

MAIN_D[15..0]

3J 065A08 KONAMI MASK 8M - 16BIT — CMOS ROM 054903

H1 065A08 KONAMI MASK 8M - 16BIT — CMOS ROM 054903

1L 065A08 KONAMI MASK 8M - 16BIT — CMOS ROM 054903

3L 065A08 KONAMI MASK 8M - 16BIT — CMOS ROM 054903

4J 4K 5J 5K 5L 4L 5M 4M — 3-state flip flops 74LS374

6K 7K 8K 11K — buffers 74LS245

9K 10K — buffers 74LS245

6HB 74LS08 — 11HD 74LS32 — 12CC 74LS32 — 6JD 74LS32

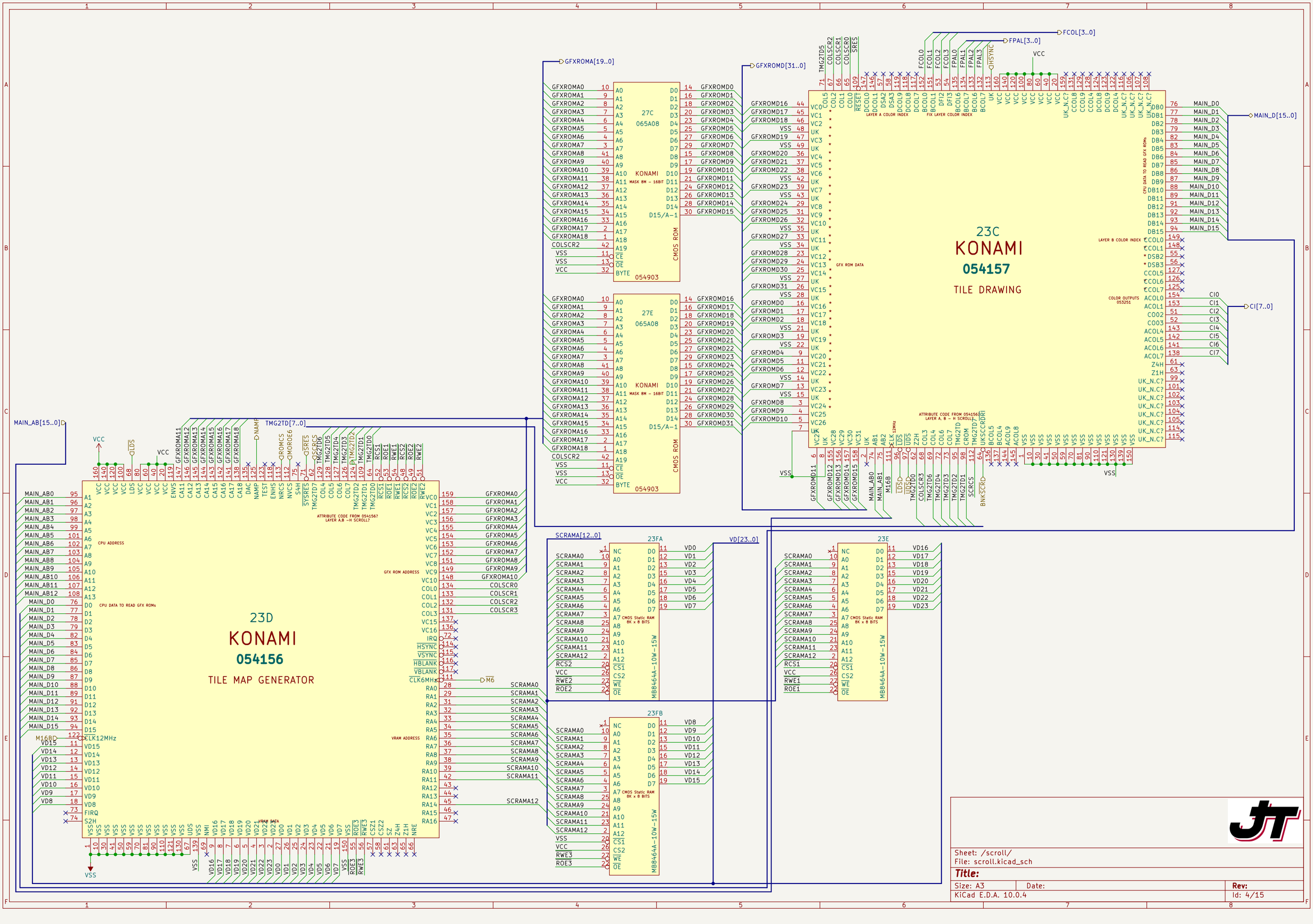
3H0 74LS04

MB8464A-80-10-15/L-LL CMOS Static RAM 8K x 8 BITS

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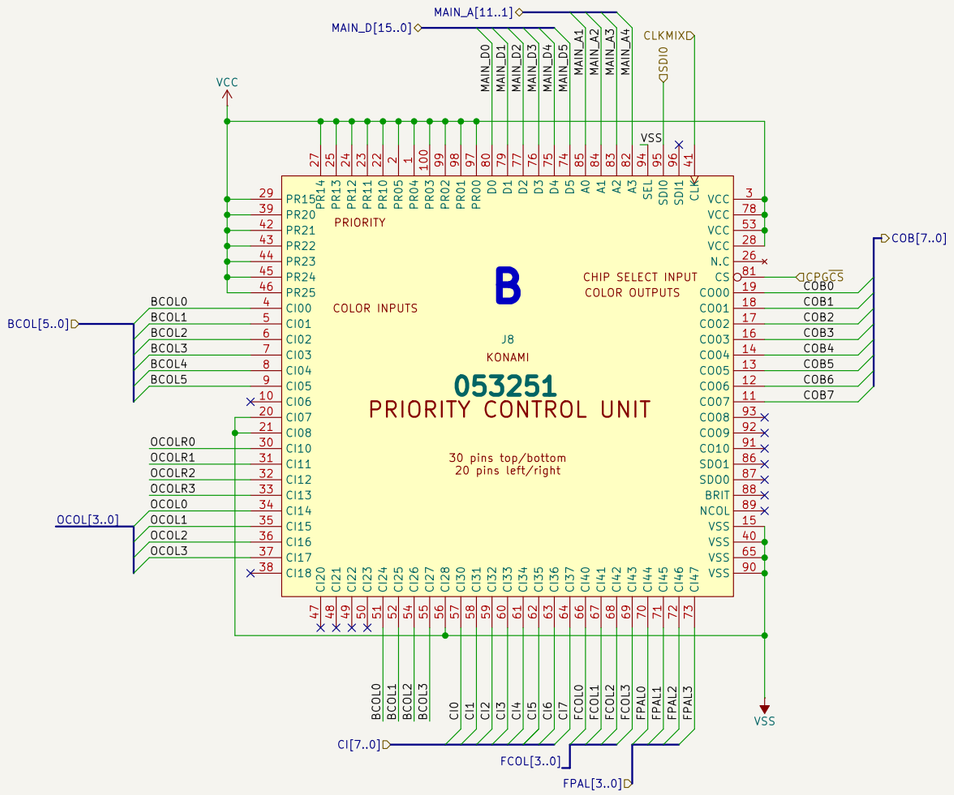
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 3/15

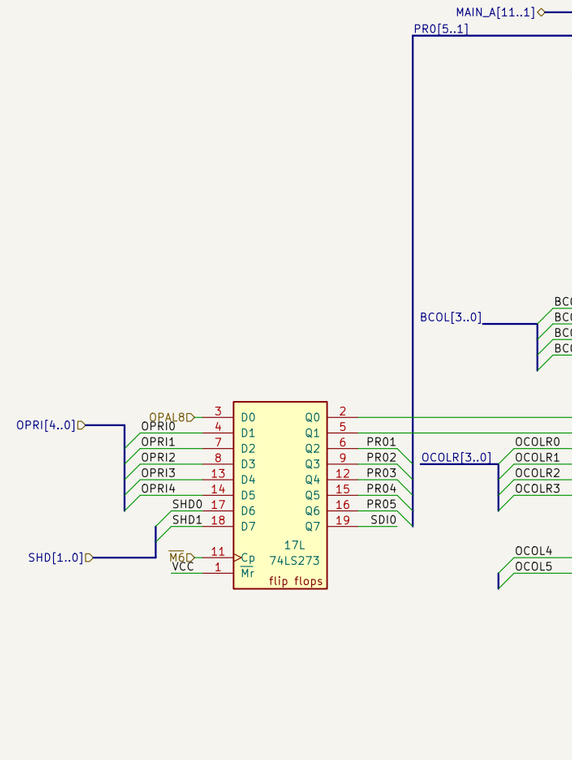


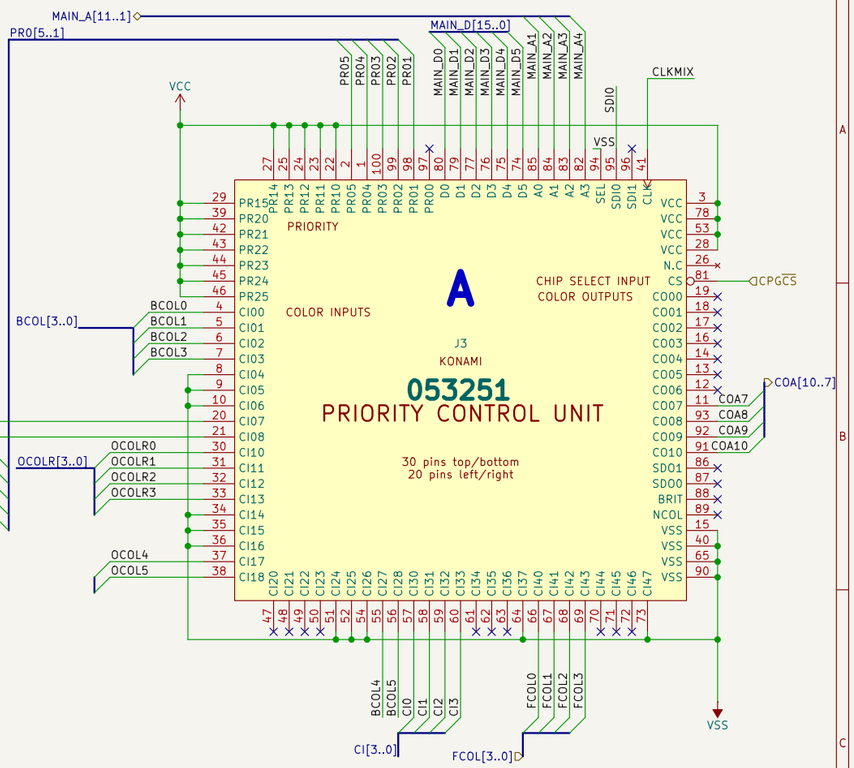
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File: scroll.kicad_sch

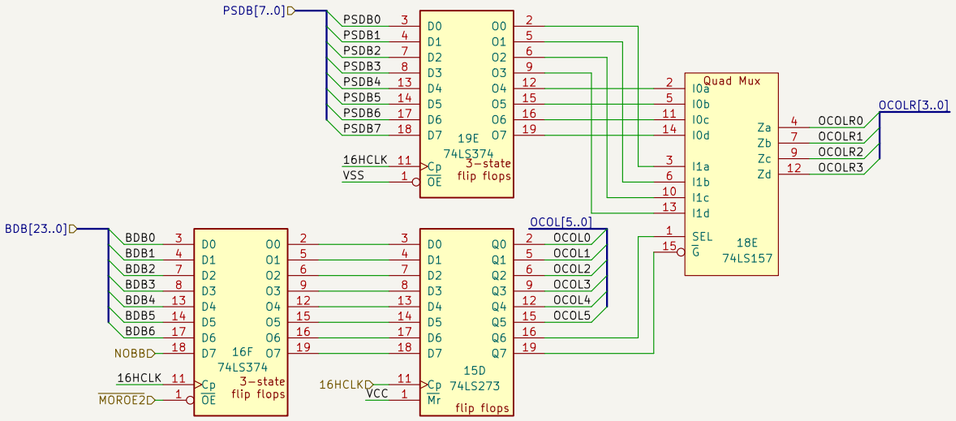
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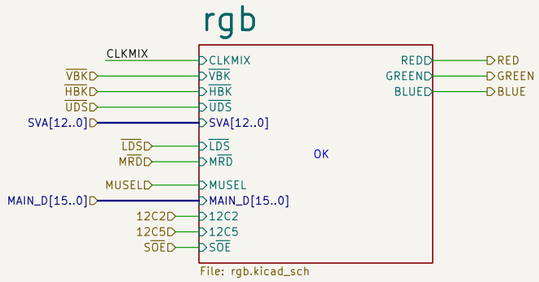
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 4/15











Sheet: /colmix/
File: colmix.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/15

RGB[15..0]

13B
74LS273

RGB0 RGB1 RGB2 RGB3 RGB4 RGB5 RGB6 RGB7 → B0 B1 B2 B3 B4 R0 R1 R2
CLKMIXD, VCC

14B
74LS273

RGB8 RGB9 RGB10 RGB11 RGB12 RGB13 RGB14 SHAD → R3 R4 G0 G1 G2 G3 G4
CLKMIX, VCC

11B
SRAM32K8B

SVA[12..0]
SVA0 SVA1 SVA2 SVA3 SVA4 SVA5 SVA6 SVA7 SVA8 SVA9 SVA10 SVA11 SVA12 VCC
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14
D0 D1 D2 D3 D4 D5 D6 D7 = RGB0 RGB1 RGB2 RGB3 RGB4 RGB5 RGB6 RGB7
VSS, SOED, LDS, 12C5
12CB 74LS32
CS OE WR, VCC VSS

10B
74LS245
A->B CE
RGB0 RGB1 RGB2 RGB3 RGB4 RGB5 RGB6 RGB7 → MAIN_D0 MAIN_D1 MAIN_D2 MAIN_D3 MAIN_D4 MAIN_D5 MAIN_D6 MAIN_D7
buffers
MRD, MUSEL

12B
SRAM32K8B

SVA[12..0]
SVA0 SVA1 SVA2 SVA3 SVA4 SVA5 SVA6 SVA7 SVA8 SVA9 SVA10 SVA11 SVA12 VCC
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 A14
D0 D1 D2 D3 D4 D5 D6 D7 = RGB8 RGB9 RGB10 RGB11 RGB12 RGB13 RGB14 SHAD
VSS, SOED, UDS, 12C2
12CA 74LS32
CS OE WR, VCC VSS

14E
74LS245
A->B CE
RGB8 RGB9 RGB10 RGB11 RGB12 RGB13 RGB14 SHAD → MAIN_D8 MAIN_D9 MAIN_D10 MAIN_D11 MAIN_D12 MAIN_D13 MAIN_D14 MAIN_D15
buffers
MRD, MUSEL

MAIN_D[15..0]

12AA 74LS07
R3 1K
6HC 74LS08
VBKD, HBKD, CLKMIX
17AB 74LS74
17AA 74LS74
R5 1K
12AF 74LS07
11AA 74LS07
R4 1K

B4 B3 B2 B1 B0 → 12AB 13AB 13AA 10AF 13AF 10AA 74LS07
G4 G3 G2 G1 G0 → 13AC 10AC 11AE 12AE 11AB 11AB 74LS07
R4 R3 R2 R1 R0 → 10AB 12AB 13AE 12AC 13AB 11AC 74LS07

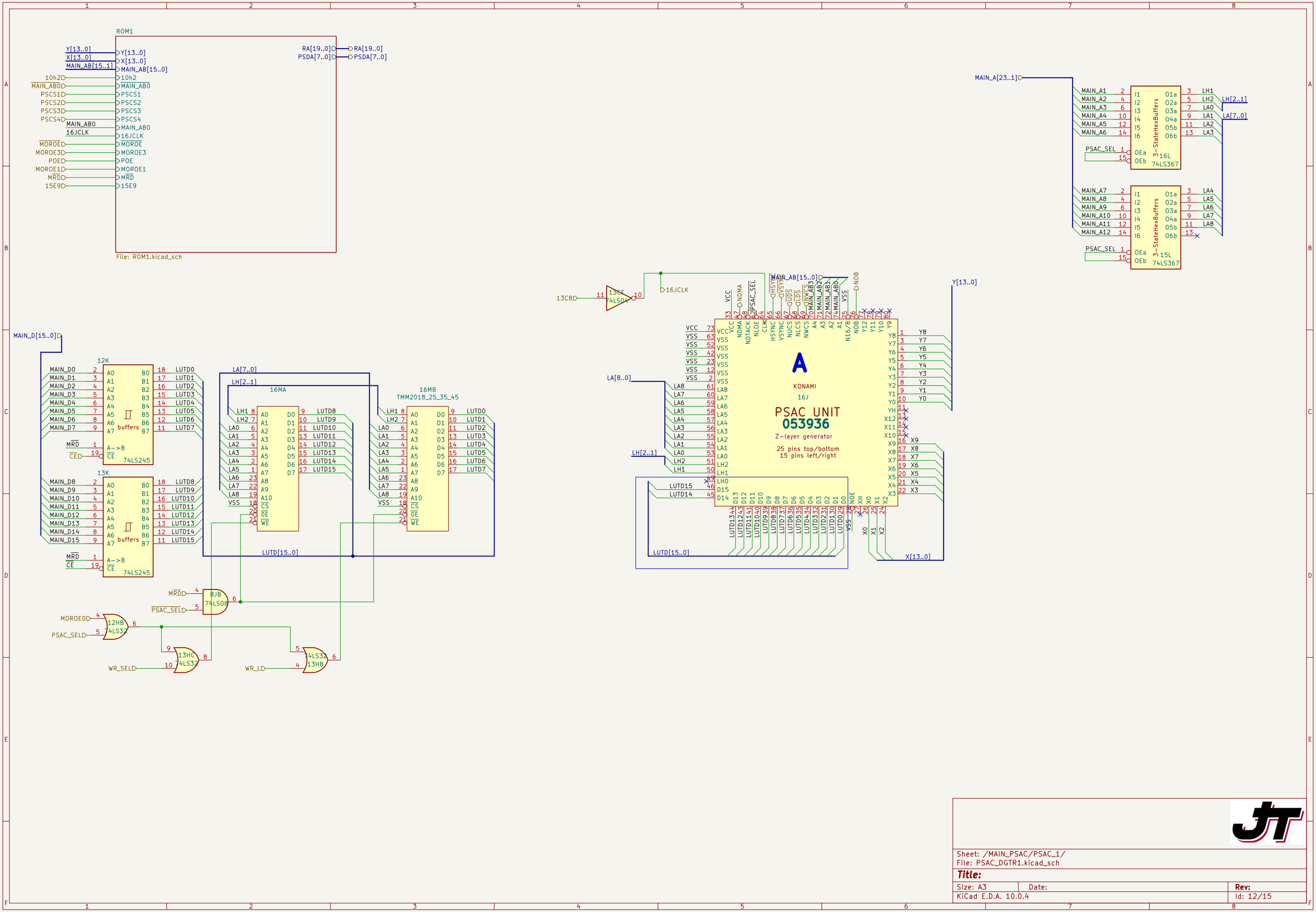
U63
U33
U134
VCC
VSS
D4* D3* D2* D1* D0* BK*
Pull Up*
DIGITAL INPUT ANALOG OUTPUT AO RGB
KONAMI 052535 5-bit Video DAC
BLUE
RED
GREEN

Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 11/15

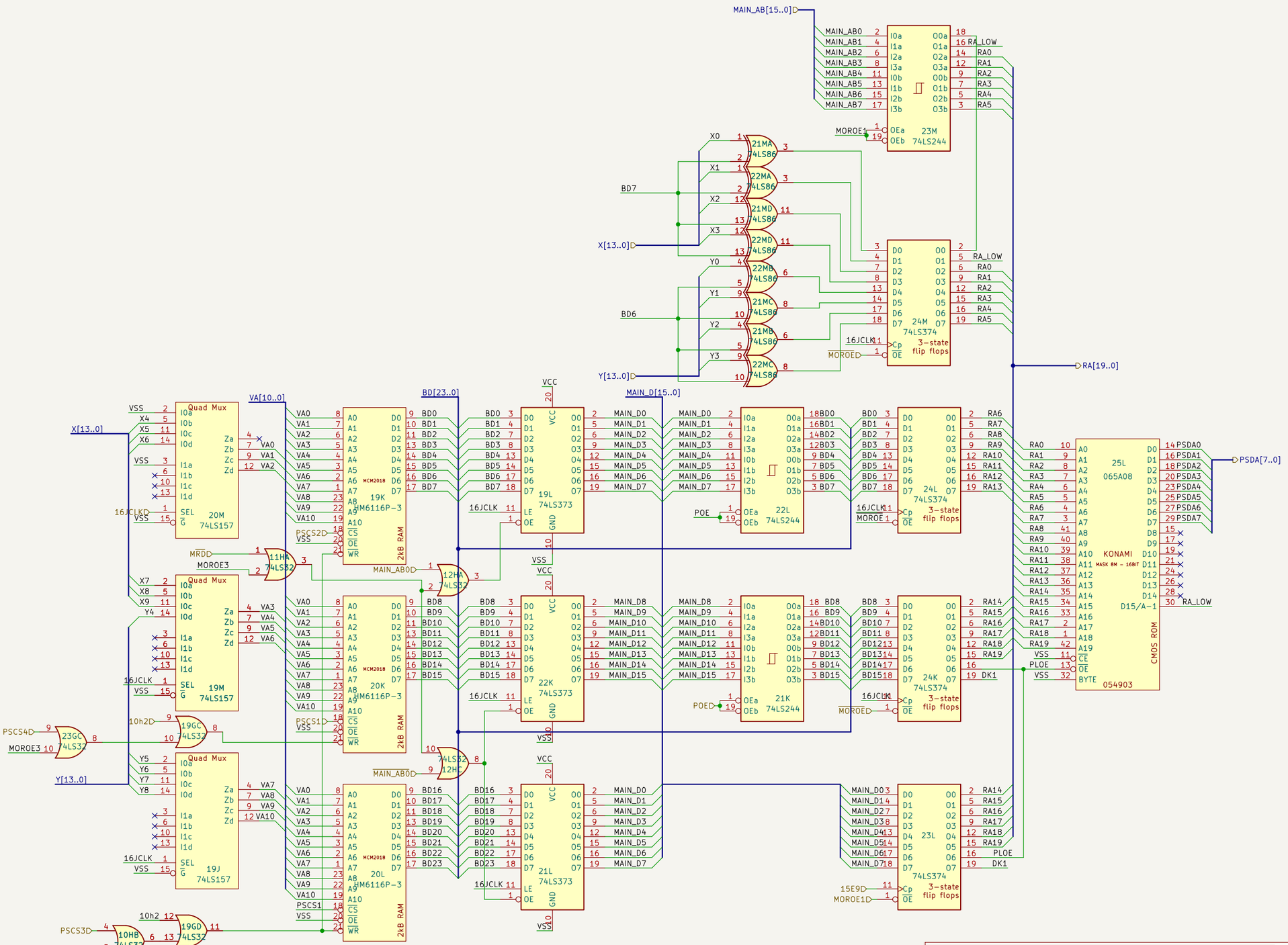
MOROE1 5 18FC 74LS04 6 MOROE1
MAIN_AB0 3 3HB 74LS04 4 MAIN_AB0
AS 1 3HA 74LS04 2 AS
MOROE2 1 18FA 74LS04 2 MOROE2
NDMA_A 5 16GC 74LS04 6
NDMA_B 9 16GB 74LS04 8
11HC 74LS32 13 12 11
R/WD 11 3GE 74LS04 10 MRD
VSS 3 4GB 74LS10 6
CLKMIX 3 3GB 74LS04 4
R/WD 1 4FA 74LS32 3 POEB
2
13 74LS32 11 POEA
5GD
12
R/WD
VCC 14 7JA 74LS74 5
D Q
VCC 2 C
AS 3 6
VCC 14 14KB 74LS74 9 PSAC_SEL
D Q
CLKMIX 11 C 8 PSAC_SEL
VCC 7
LDS 12 10HD 74LS32 11
R/WD 13
UDS 9 10HC 74LS32 8
R/WD 10
WR_SEL
16HCLK 1
VSS 15
VSS 2 Quad Mux
VSS 5 I0a I0b I0c I0d Za Zb Zc Zd 4 PSCS1
VCC 11 7 PSCS2
VCC 14 I1a I1b I1c I1d 9 PSCS3
MAIN_AB0 3 12 PSCS4
MAIN_AB0 6
WR_L 10
13
SEL
G 24G
74LS157
MAIN_D[7..0]
MAIN_D0 2 I0a I1a I2a I3a I0b I1b I2b I3b 18 BDB0
MAIN_D1 4 16 BDB1
MAIN_D2 6 14 BDB2
MAIN_D3 8 12 BDB3
MAIN_D4 11 9 BDB4
MAIN_D5 13 7 BDB5
MAIN_D6 15 5 BDB6
MAIN_D7 17 3 BDB7
POEB 1 OEa OEb 15F 74LS244
19
BDB[23..0]
MAIN_A[23..1]
AS 1 VCC 20 9H PAL16L8
MAIN_A23 2 I1 I2 I3 I4 I5 I6 I7 I8 I9 I10 VCC GND I01 I02 I03 I04 I05 I06 I07 I08 19 MOROE0
MAIN_A22 3 18 MOROE1
MAIN_A21 4 17 MOROE3
MAIN_A20 5 16 MOROE4
MAIN_A19 6 15 MOROE5
MAIN_A18 7 14 MOROE6
MAIN_A17 8 13 MOROE7
MAIN_A16 9 12
MAIN_A15 11
VSS 10
MOROE0 13 74LS08 8JD 11 8JP11
NWCS_AD 12
PSAC_SEL
VCC
AS 1 20 6JB 74LS32 6 CEA
MAIN_A23 2 I1 I2 I3 I4 I5 I6 I7 I8 I9 I10 VCC GND I01 I02 I03 I04 I05 I06 I07 I08 19 8H19
MAIN_A22 3 18 MPLXH
MAIN_A21 4 17 MUSEL
MAIN_A20 5 16 PAL8HP16
MAIN_A19 6 15
MAIN_A18 7 14
MAIN_A17 8 13 8H13
MAIN_A16 9 12
MAIN_A15 11
VSS 10
MAIN_A[23..1]
8H PAL16L8
4FB 74LS32 6 4F6
MOROE7 1 6HA 74LS08 3 9 4FC 74LS32 8CEB
NWCS_BD 2 10
12HD 74LS32 11 PSOE
MRD 13
WR_L 1 13HA 74LS32 3 PSWR2
2
13 13HD 74LS32 11 PSWR1
WR_SEL 12
COB[7..0]
MAIN_AB[15..0]
MAIN_AB0 2 Quad Mux
MAIN_AB1 5 I0a I0b I0c I0d Za Zb Zc Zd 4 SVA0
MAIN_AB2 11 7 SVA1
MAIN_AB3 14 I1a I1b I1c I1d 9 SVA2
COB0 3 12 SVA3
COB1 6
COB2 10
COB3 13
MUSELD 1 SEL
VSS 15 G 15B1
74LS157
MAIN_AB4 2 Quad Mux
MAIN_AB5 5 I0a I0b I0c I0d Za Zb Zc Zd 4 SVA4
MAIN_AB6 11 7 SVA5
MAIN_AB7 14 I1a I1b I1c I1d 9 SVA6
COB4 3 12 SVA7
COB5 6
COB6 10
COB7 13
MUSEL 1 SEL
VSS 15 G 22B
74LS157
SVA[12..0]
MAIN_AB8 2 Quad Mux
MAIN_AB9 5 I0a I0b I0c I0d Za Zb Zc Zd 4 SVA8
MAIN_AB10 11 7 SVA9
MAIN_AB11 14 I1a I1b I1c I1d 9 SVA10
COA7 3 12 SVA11
COA8 6
COA9 10
COA10 13
COA[10..7]
MUSEL 1 SEL
VSS 15 G 16C
74LS157
MAIN_AB13 2 Quad Mux
MRD 5 I0a I0b I0c I0d Za Zb Zc Zd 4 SVA12
WR_L 11 7
WR_SEL 14 9 SOE
12 12C5
10C9 3 I1a I1b I1c I1d
VSS 6
VCC 10
VCC 13
MUSEL 1 SEL
VSS 15 G 14C
74LS157
MAIN_D0 2 I0a I1a I2a I3a I0b I1b I2b I3b 18 3
MAIN_D1 4 16 4
MAIN_D2 6 14 7
MAIN_D3 8 12 8
MAIN_D4 11 9 13
MAIN_D5 13 7 14
MAIN_D6 15 5 VSS 17
MAIN_D7 17 3 NOBAD 18
POEAD 1 OEa OEb 21G 74LS244
19
16JCLK 11 D0 D1 D2 D3 D4 D5 D6 D7 17E 74LS374 O0 O1 O2 O3 O4 O5 O6 O7 3-state flip flops
MOROE1 1
OE
16JCLK 11 D0 D1 D2 D3 D4 D5 D6 D7 17D 74LS273 Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7 2 BCOL0
VCC 1 Cp 5 BCOL1
Mr 6 BCOL2
9 BCOL3
12 BCOL4
15 BCOL5
16
19
flip flops
BCOL[5..0]
PSDB[7..0]
PSDB0 2 I0a I1a I2a I3a I0b I1b I2b I3b 18 MAIN_D0
PSDB1 4 16 MAIN_D1
PSDB2 6 14 MAIN_D2
PSDB3 8 12 MAIN_D3
PSDB4 11 9 MAIN_D4
PSDB5 13 7 MAIN_D5
PSDB6 15 5 MAIN_D6
PSDB7 17 3 MAIN_D7
MOROE1D 1 OEa OEb 20G 74LS244
19
MAIN_D[15..0]
PSDB0 2 I0a I1a I2a I3a I0b I1b I2b I3b 18 MAIN_D0
PSDB1 4 16 MAIN_D1
PSDB2 6 14 MAIN_D2
PSDB3 8 12 MAIN_D3
PSDB4 11 9 MAIN_D4
PSDB5 13 7 MAIN_D5
PSDB6 15 5 MAIN_D6
PSDB7 17 3 MAIN_D7
MOROE2D 1 OEa OEb 19F 74LS244
19
RA[13..6]
MAIN_AB15..8]
MAIN_AB8 2 I0a I1a I2a I3a I0b I1b I2b I3b 18 RA6
MAIN_AB9 4 16 RA7
MAIN_AB10 6 14 RA8
MAIN_AB11 8 12 RA9
MAIN_AB12 11 9 RA10
MAIN_AB13 13 7 RA11
MAIN_AB14 15 5 RA12
MAIN_AB15 17 3 RA13
MOROE1D 1 OEa OEb 23K 74LS244
19
RB[13..6]
MAIN_AB8 2 I0a I1a I2a I3a I0b I1b I2b I3b 18 RB6
MAIN_AB9 4 16 RB7
MAIN_AB10 6 14 RB8
MAIN_AB11 8 12 RB9
MAIN_AB12 11 9 RB10
MAIN_AB13 13 7 RB11
MAIN_AB14 15 5 RB12
MAIN_AB15 17 3 RB13
MOROE2 1 OEa OEb 27G 74LS244
19
PSAC_1
MAIN_A[23..1] MAIN_A[23..1] RA[19..0] RA[19..0]
MAIN_D[15..0] MAIN_D[15..0] PSDA[7..0] PSDA[7..0]
MAIN_AB[15..0] MAIN_AB[15..0] NOB NOBA
16JCLK 16JCLK
NDMA NDMA_A
MRD MRD
LDS LDS
UDS UDS
HSYNC HSYNC
NWCS_A NWCS
VSYNC VSYNC
MAIN_AB0 MAIN_AB0
CEA CE
PSCS1 PSCS1
PSCS2 PSCS2
PSCS3 PSCS3
PSCS4 PSCS4
13C8 13C8
10h2 10h2
PSAC_SEL PSAC_SEL
PSAC_SEL PSAC_SEL
WR_SEL WR_SEL
WR_L WR_L
POEA POE
MOROE1 MOROE1
MOROE3 MOROE3
MOROE1 MOROE
MOROE0 MOROE0
15E9 15E9
File: PSAC_DGTR1.kicad_sch
PSAC_2
MAIN_A[23..1] MAIN_A[23..1] RB[19..0] RB[19..0]
MAIN_D[15..0] MAIN_D[15..0] PSDB[7..0] PSDB[7..0]
MAIN_AB[15..0] MAIN_AB[15..0] NOB NOBB
16HCLK 16HCLK
NDMA NDMA_B
BDB[23..0] BDB[23..0]
MRD MRD
LDS LDS
UDS UDS
HSYNCD HSYNC
NWCS_B NWCS
VSYNC VSYNC
MAIN_AB0 MAIN_AB0
CEB CE
PSCS1 PSCS1
PSCS2 PSCS2
PSCS3 PSCS3
PSCS4 PSCS4
13C8 13C8
10h2 10h2
PSAC_SEL PSAC_SEL
PSAC_SEL PSAC_SEL
WR_SEL WR_SEL
WR_L WR_L
POEB POE
MOROE7 MOROE7
MOROE4 MOROE4
MOROE2 MOROE2
MOROE2 MOROE2
15E9 15E9
File: PSAC_DGTR2.kicad_sch
Sheet: /MAIN_PSAC/
File: PSAC.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 6/15



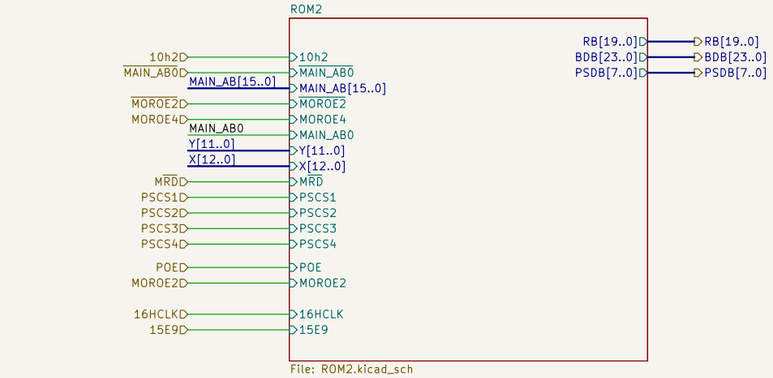
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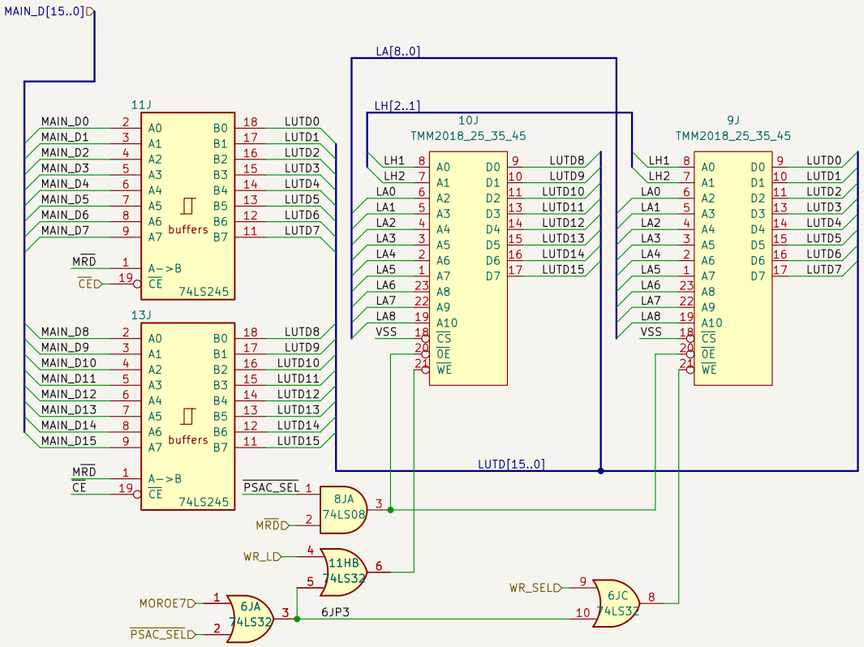
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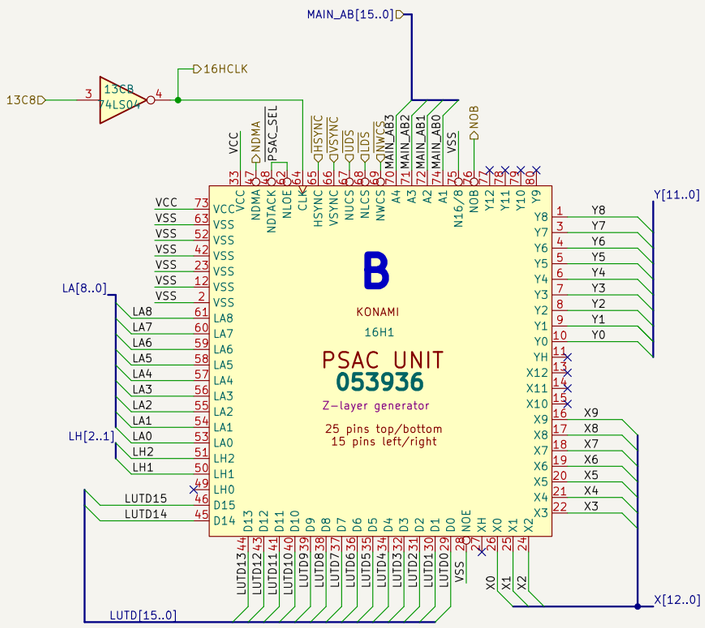
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KiCad E.D.A. 10.0.4		Id: 12/15

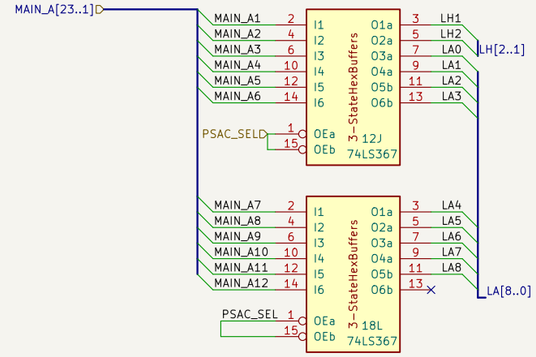


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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 14/15

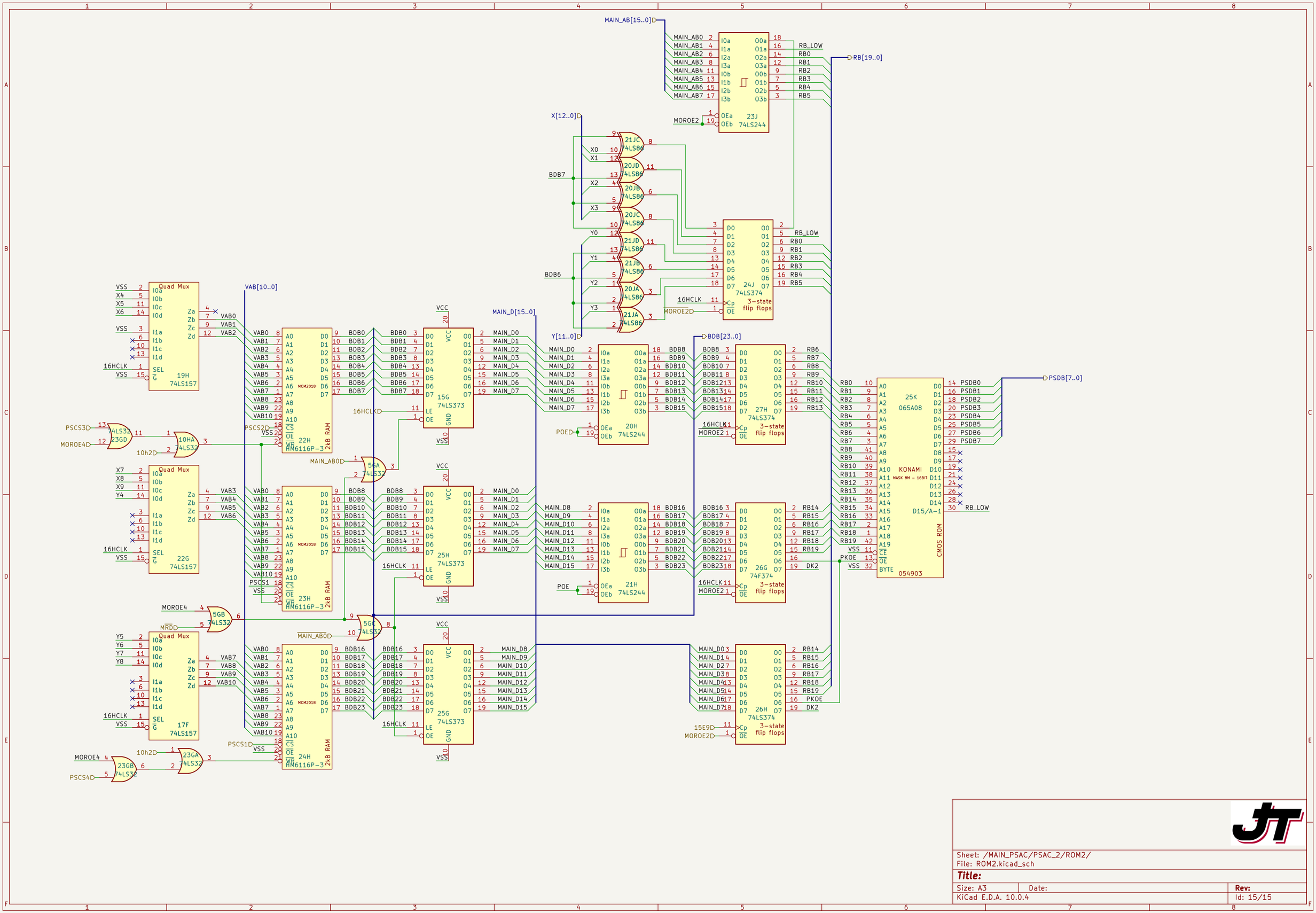








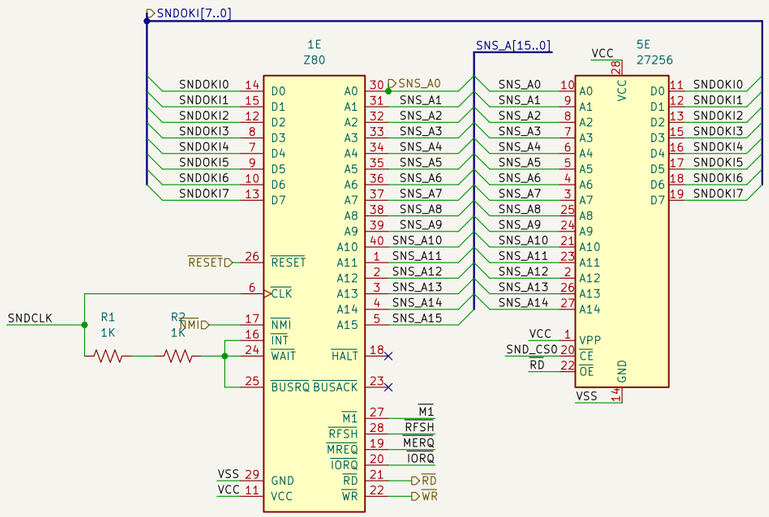
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KiCad E.D.A. 10.0.4 Id: 13/15

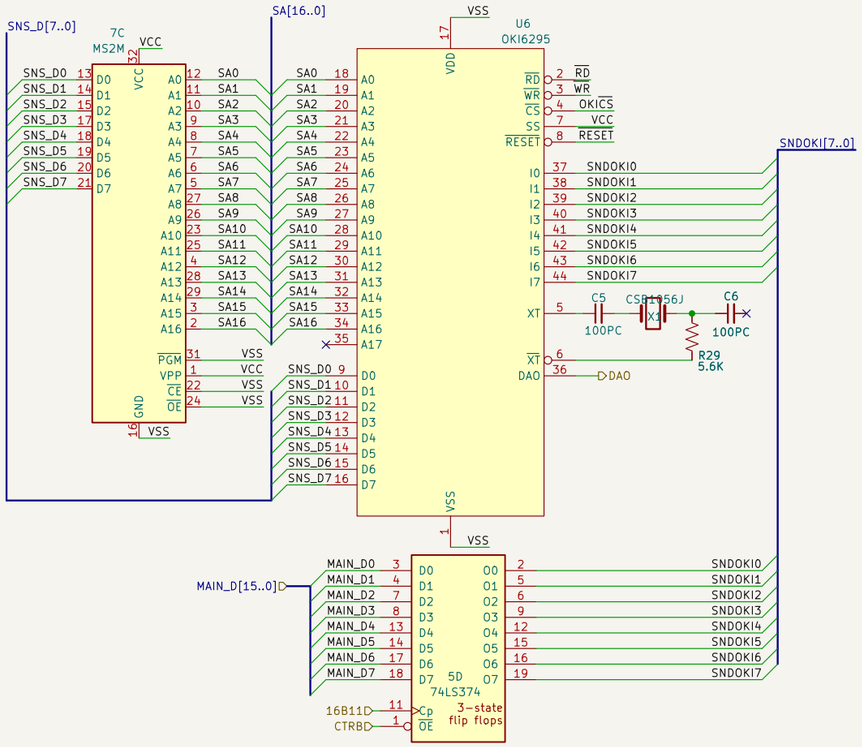


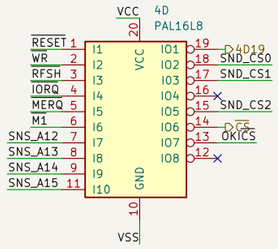
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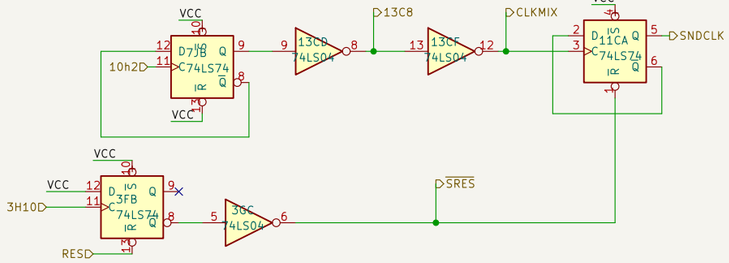
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KiCad E.D.A. 10.0.4		Id: 15/15

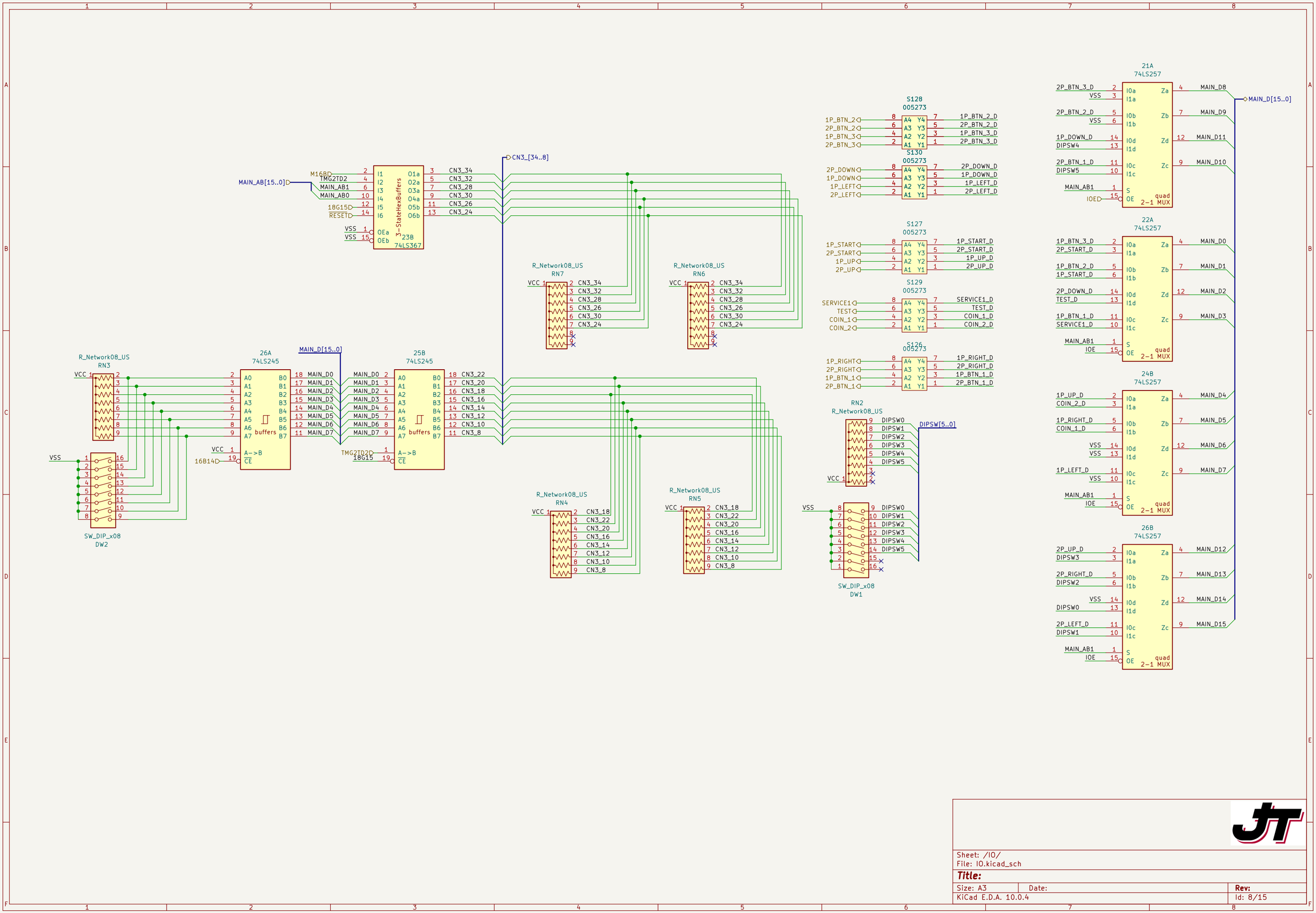




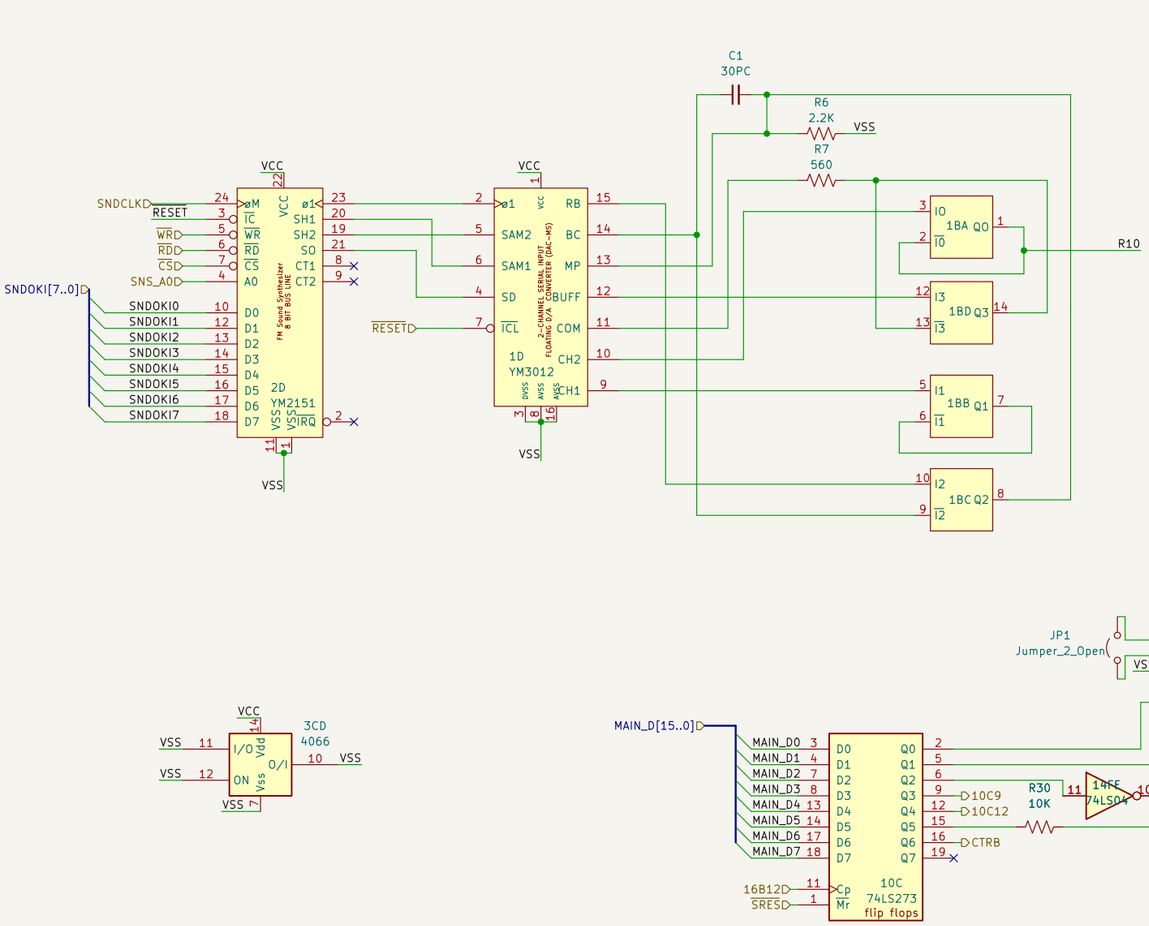


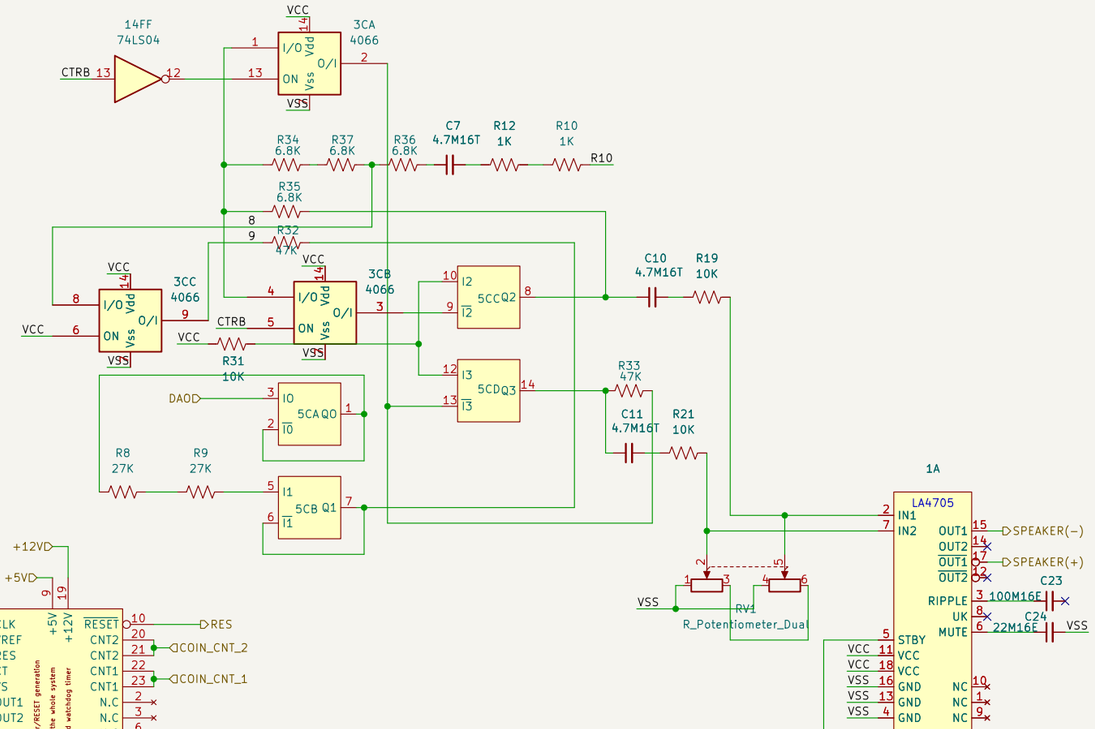


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KiCad E.D.A. 10.0.4		Id: 7/15



Sheet: /IO/		
File: IO.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 8/15





Banpresto/Konami

Sheet: /analog/

File: analog.kicad_sch

Title: DBZ

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 9/15

VCC

VSS

11AG 74LS07 | 3GG 74LS04 | 13CG 74LS04 | 22JG 74LS04 | 8JE 74LS08 | 11CC 74LS74 | 6JE 74LS32 | 12CE 74LS32 | 11HE 74LS32 | 6HE 74LS08 | 3HG 74LS04 | 18FG 74LS04 | 14KC 74LS74 | 10HE 74LS32 | 5GE 74LS32 | 4FE 74LS32 | 13HE 74LS32 | 12HE 74LS32 | 7HC 74LS74 | 5HC 74LS74 | 4HC 74LS74 | 2GC 74LS21 | 16GG 74LS04 | 7JC 74LS74 | 3FC 74LS74 | 14FG 74LS04

5CE | 1BE | 19GE 74LS32 | 21JE 74LS86 | 20JE 74LS86 | 23GE 74LS32 | 21ME 74LS86 | 22ME 74LS86 | 12AG 74LS07 | 13AG 74LS07 | 10AG 74LS07 | 4GD 74LS10 | 17AC 74LS74

USED IN DBZ2

RB[19..0] → 27K 065A08 (KONAMI, MASK 8M – 16BIT, CMOS ROM) 054903 → PSDB[7..0]

RA[19..0] → TO UNPOPULATED

27L 065A08 (KONAMI, MASK 8M – 16BIT, CMOS ROM) 054903 → PSDA[7..0]

MAIN_A[23..1] → EPROM 4 Mbit 512 KB

CN3_[34..8] → CN3 Conn_02x20_Odd_Even

4GA 74LS10 | 4GC 74LS10 | 4FD 74LS32

10AB 74LS07 | 10AE 74LS07 | 19GA 74LS32 | 19GB 74LS32

16GE 74LS04 | 22JA 74LS04 | 22JB 74LS04 | 22JE 74LS04 | 12CD 74LS32

22JF 74LS04 | 18FF 74LS04 | 3HC 74LS04 | 13CA 74LS04 | 14FA 74LS04

D14KA 74LS74 | D7HA 74LS74

2GB 74LS21

Sheet: /unpopulated/	
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Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Id: 10/15